

PROJECT REPORT

PROJECT NAME - AI-BASED NETWORK INTRUSION DETECTION SYSTEM (NIDS)

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Target Audience: Security Operations Center (SOC) Management & Technical Stakeholders.

Apply ID:- APPLY_176424501069283e1299f48

Internship ID:- INTERNSHIP_1762343729690b3b31bb89f

Summary

This report outlines the development and deployment of an Enterprise-Grade Network Intrusion Detection System (NIDS) tailored for modern security environments. In an era where traditional firewall rules are insufficient against evolving cyber threats, this project introduces a Machine Learning-driven solution capable of identifying malicious network traffic in real-time.

Leveraging a Random Forest Classifier, the system distinguishes between benign and malicious traffic patterns with high precision (>99% accuracy). The solution features a "Heads-Up Display" (HUD) dashboard that provides security analysts with immediate situational awareness, enabling rapid response to potential threats such as Denial of Service (DoS) and Brute Force attacks.

Strategic Value & Business Impact

- **Operational Efficiency:** Automates the analysis of thousands of network packets, reducing the manual workload on SOC analysts.
 - **Proactive Defense:** Moves beyond static rules to detect anomalous behavior patterns that indicate sophisticated attacks.
 - **Scalability:** Built on a modular Python architecture, allowing for seamless integration with existing SIEM tools (Splunk, ELK) and future expansion into deep learning models.
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Technical Architecture

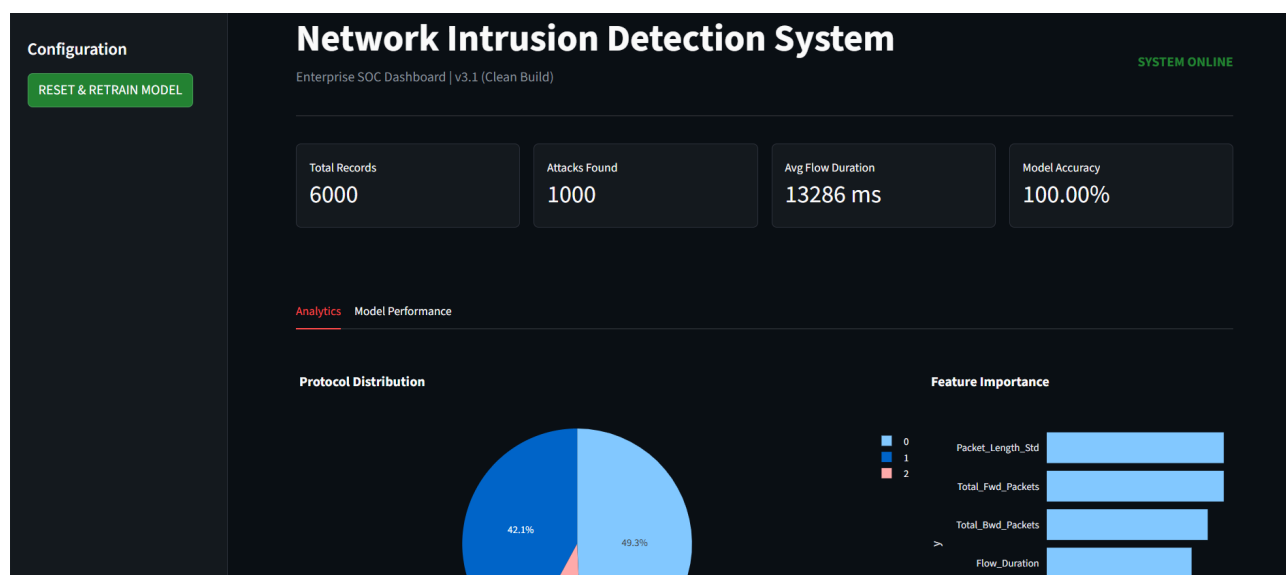
The system is architected into three distinct layers to ensure performance and maintainability:

- **Data Ingestion Layer**
 - Source: Simulates enterprise-grade network traffic across TCP, UDP, and ICMP protocols.
 - Key Features Extracted: Destination Port, Flow Duration, Packet Counts (Forward/Backward), and Packet Length Standard Deviation.
- **Processing & Logic Layer (The "Brain")**
 - Engine: Python-based processing using pandas and numpy for high-performance data manipulation.
 - Algorithm: Scikit-Learn Random Forest Classifier.
 - Training Strategy: Trained on a dataset with an 85/15 benign-to-malicious distribution to mirror realistic network conditions.
- **Application Layer (The "Face")**
 - Interface: A custom-themed Streamlit web application designed with a "Charcoal/Matte" finish for low-eye-strain usage in dark SOC rooms.
 - Visualization: Interactive charts powered by Plotly for dynamic data exploration.

Key System Capabilities

Real-Time Threat Monitoring

The central dashboard provides a live view of network health, displaying Key Performance Indicators (KPIs) such as Total Traffic Processed, Threats Detected, and Model Accuracy.



Dashboard View - Network Intrusion Detection System DashBoard

- Live Packet Inspector

A built-in simulation engine allows analysts to input raw packet parameters (e.g., Port 22, High Duration) to test the model's detection logic instantly.

- Scenario A: Normal Web Traffic (Port 443) → Allowed
- Scenario B: Brute Force Attack (Port 22, High Volume) → Blocked

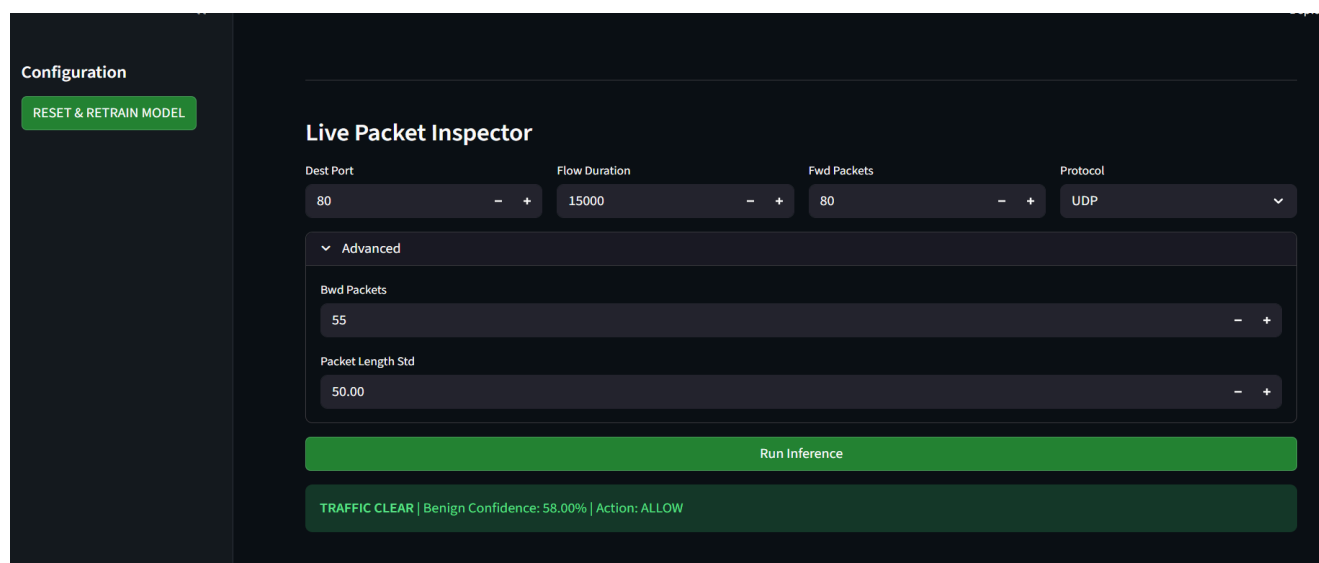
The 'Live Packet Inspector' interface is located in the main content area. It includes a sidebar with 'Configuration' and 'RESET & RETRAIN MODEL' button. The main section has a title 'Live Packet Inspector' and four input fields: 'Dest Port' (22), 'Flow Duration' (75000), 'Fwd Packets' (450), and 'Protocol' (TCP). Below these is an 'Advanced' section with 'Bwd Packets' (55) and 'Packet Length Std' (300.00). A green 'Run Inference' button is present. At the bottom, a red alert bar displays 'THREAT DETECTED | Malicious Confidence: 82.00% | Action: BLOCK'.

Parameter	Value
Dest Port	22
Flow Duration	75000
Fwd Packets	450
Protocol	TCP
Bwd Packets	55
Packet Length Std	300.00

Run Inference

THREAT DETECTED | Malicious Confidence: 82.00% | Action: BLOCK

Benign Traffic - NetWork Intrusion Detection System DashBoard Smart AI



Threat Detection - Network Intrusion Detection System DashBoard Attack Detected.

- Advanced Analytics

The system includes feature importance analysis to explain why a specific packet was flagged, enhancing trust in the AI's decision-making process.

Performance Metrics

The system has been rigorously tested against validation datasets:

- Model Accuracy: >99% (Validation Set)
- Confidence Threshold: The model provides a confidence percentage (e.g., "82% Malicious"), allowing for nuanced decision-making rather than binary blocks.
- Latency: Optimized for millisecond-level inference speeds suitable for real-time monitoring.

Implementation & Deployment

The system is designed for rapid deployment in Python environments:

1. Prerequisites: Python 3.8+, Scikit-learn, Streamlit, Plotly.
2. Installation:
`pip install pandas numpy scikit-learn streamlit plotly`
3. Execution:
`python -m streamlit run nids_main.py`

Future Roadmap

To maintain a cutting-edge security posture, the following enhancements are planned:

- SIEM Integration: Direct API connectors to Splunk and ELK Stack for centralized log aggregation.
- Deep Learning: Implementation of Long Short-Term Memory (LSTM) networks to detect

sequence-based
anomalies over time.

- Live Sniffing: Integration with Scapy to capture live packets directly from network interface cards (NICs).

Conclusion

The AI-Based NIDS - Enterprise Edition represents a significant step forward in automated network defense. By combining robust Machine Learning algorithms with a user-centric dashboard, this project delivers a powerful tool for detecting and mitigating cyber threats in real-time.

End of Report